Javier A. Cuevas Chabrier

PROFESSIONAL PORTFOLIO

PROJECTS PRESENTATION

WEBSITE: JAVIER A. CUEVAS CHABRIER



About Me

Interests/Hobbies:

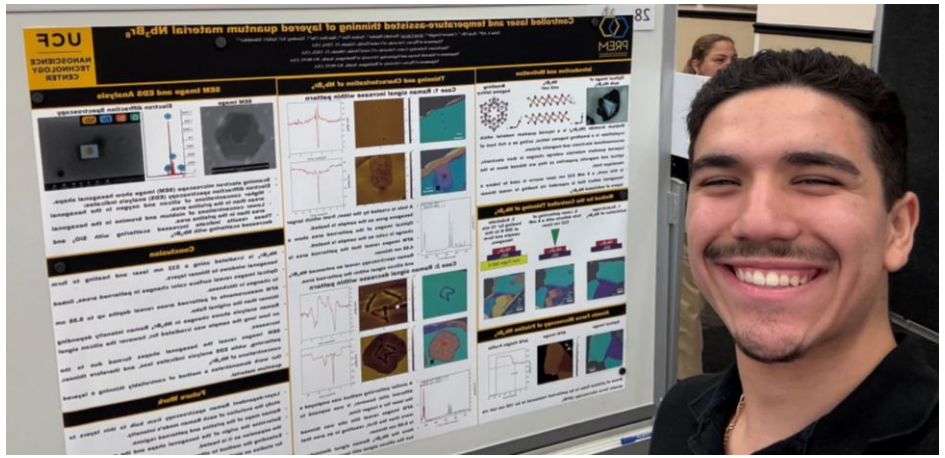
- I love baseball, volleyball, and rock climbing.
- I like Chinese food.
- I like to learn and apply myself to different challenges.
- I like to code in different languages (C++, Python, etc.).
- Interested in applying AI to engineering applications, building propulsion and avionics systems, and conducting research and development work to pioneer new aerospace technologies.

Plans:

- Obtain a B.S. in Mechanical Engineering and a B.S. in Computer Science by 2029.
- Ph.D. in Mechanical Engineering post-undergrad.

Timeline





Currently:

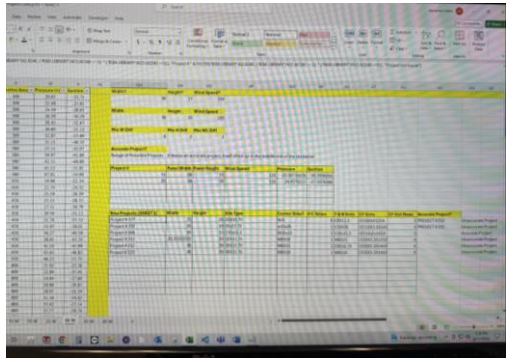
- Studying at the University of Central Florida, working to obtain two bachelor's degrees.
- Working for Lockheed Martin as a Systems and Software Engineer and conducting research at the Propulsion and Energy Research Lab.
- Involved in the Society of Hispanic Professional Engineers, Knights Experimental Rocketry, and Knights Racing BAJA SAE.

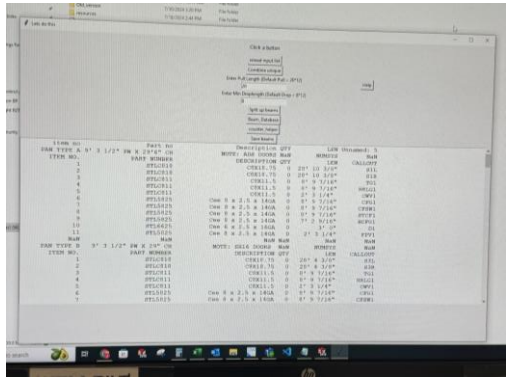
Previously:

- Studied perovskites at FSU as a research assistant.
- Worked for Well Bilt Industries as an engineering intern.
- Conducted research in Quantum Materials
- 5+ years of CAD experience.

Experience | Well Bilt Industries

- Developed Python program that would organize Excel sheets with beam data for faster manufacturing purposes and appropriate shipping (cut list management).
- Created Excel sheet that would make calculations for general doors and locate previous doors that would meet user parameters.
  - Outputs two tables: general dimensions and dimensions based on previous projects.
  - Determines what door is most accurate to user input.
- Utilized Risa – Structural Analysis to determine what beams were most appropriate for each project after calculating wind loads using Excel Sheet.
- Leveraged SolidWorks to complete drawings of hangar doors and their respective parts. Handled large assemblies, sub-assemblies, and configurations to create drawings before and after client approval.









Experience | Code Meets Bagel

- Designed, developed, and integrated an AI-powered Slack bot using Python, OpenAI, and Anthropic APIs to automate key marketing workflows and improve team communication.
- Integrated GitHub workflows to manage version control and continuous deployment, ensuring a smooth development process and scalability.
- Strengthened my experience in API integration, automation, and AI-assisted task management within collaborative environments.



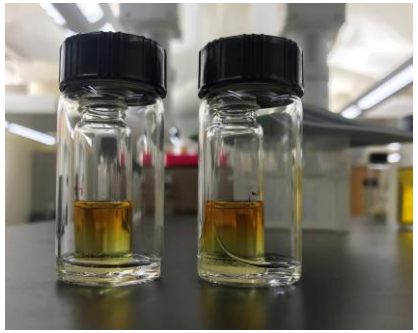
Experience | Lockheed Martin

- Working under the University of Central Florida as a CWEP contractor for Lockheed Martin.
- Develop Python-based plugins used within Cameo Systems Modeler.
- Use Cameo Systems Modeler to model client projects using plugins.
- Work alongside System and Software Engineers to automate time-consuming tasks within Cameo through the use of Plugins.
- Read and interpreted C++ plugins for identifying specified Inputs, Outputs, Publications, and Subscriptions parameters within projects.

Experience | FSU YSP





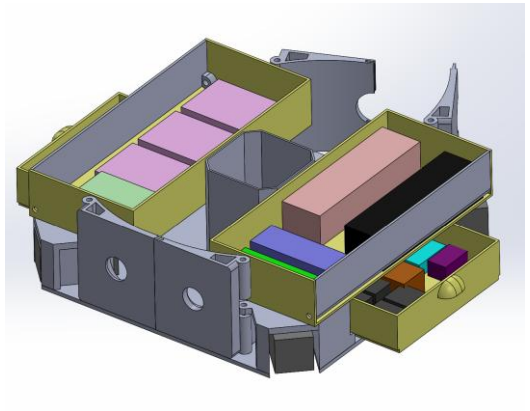


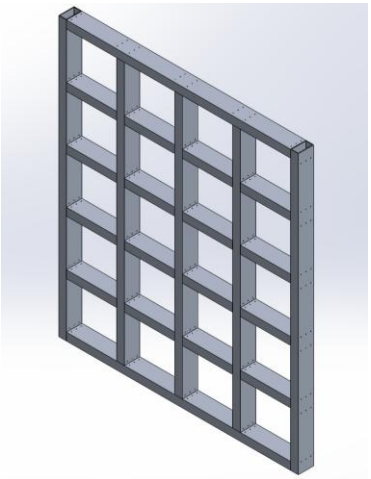


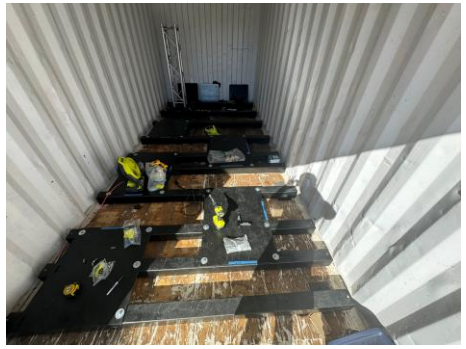


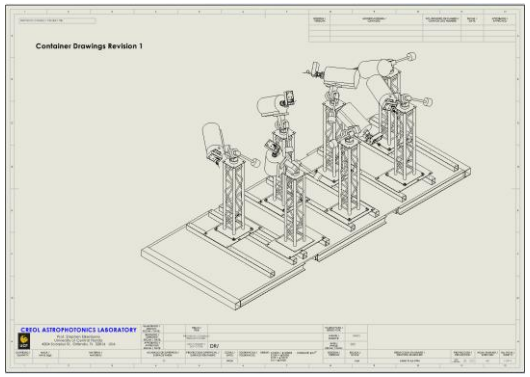
- Worked under Professor Biwu Ma of the Biochemistry Department to learn about perovskites and their applications to light Emitting Diodes (LED's).
- Learned basic concepts of materials science/materials engineering/chemical engineering that assisted in the understanding of intermolecular forces.
- Presented my thesis and conclusion to other Professors and Graduate students.

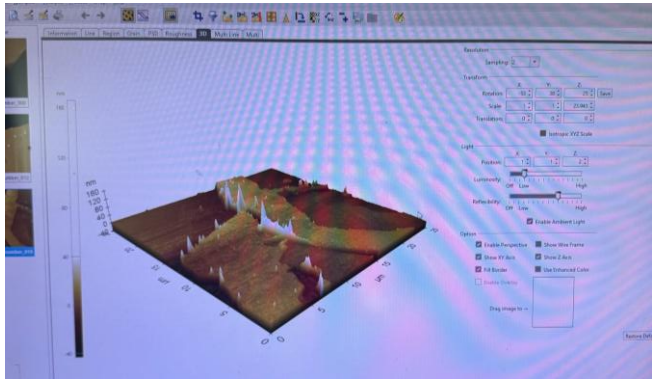
Experience | UCF Research





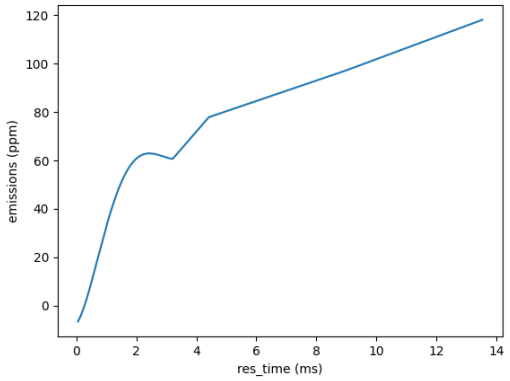


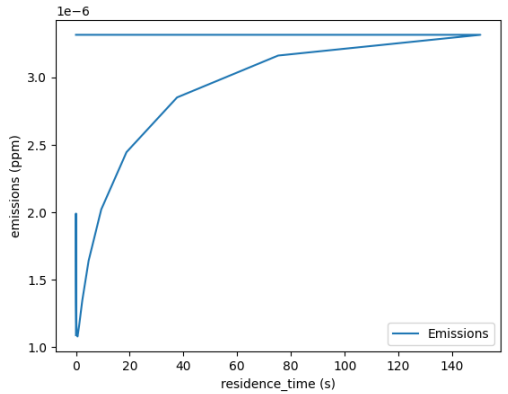




- Working under Professor Eikenberry of the Astrophysics/Optics and Photonics department to develop a Telescoping assembly for studying the solar flares of stars.
- Working under Professor Elgohary of the Aerospace engineering Department to study orbital path efficiency using ROSS.
- Working under Professor Khondaker of the Physics department, studying quantum materials and making quantum devices.

Experience | UCF Research



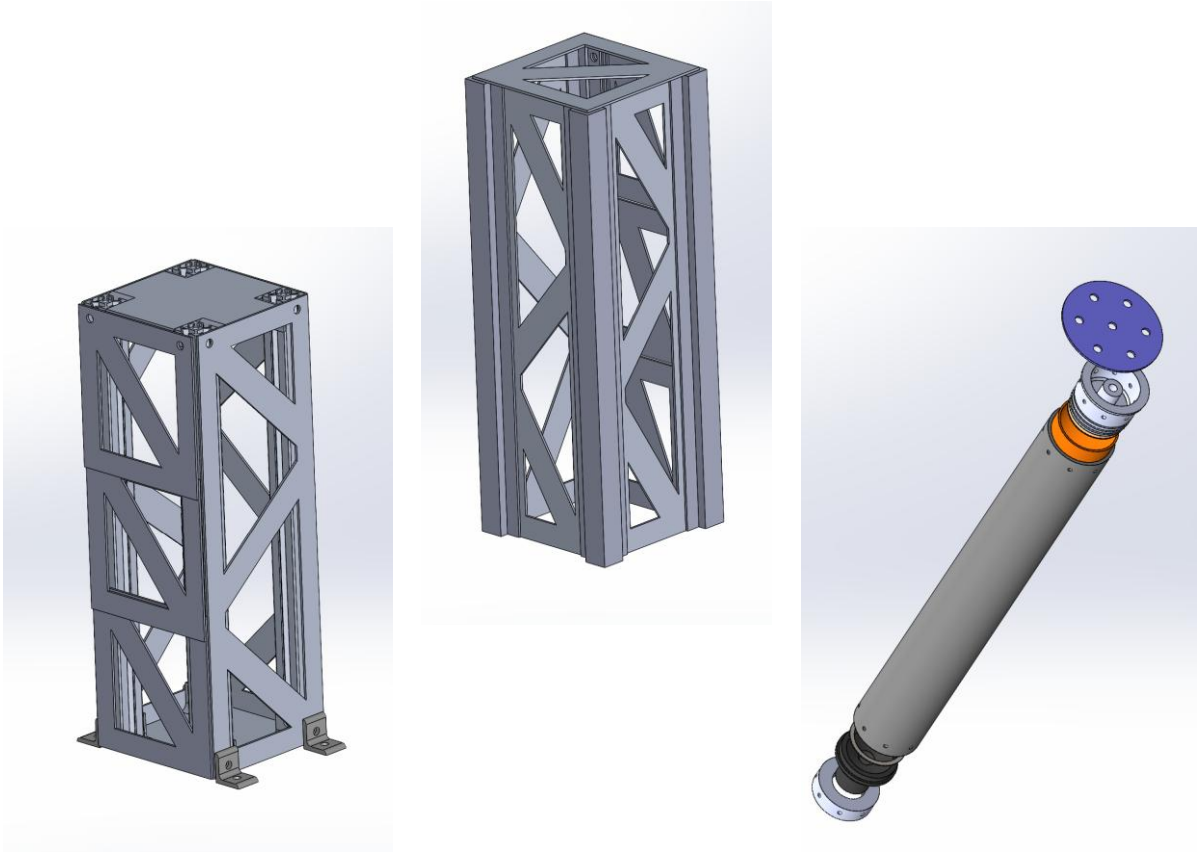


Input Type	CH4	H2	axial air	axial air.1
fuel/oxidizer	Fuel	Fuel	Oxidizer	Oxidizer
mass flow	0.000829	0.000104	0.03018	0.03018
choke diam (in)	0.032	0.019	0.113	0.113
# of orfices	1	1	1	1
discharge	0.867	0.87	0.93	0.93
R	518.2785	4124.483	286.9	286.9
Gamma	1.32	1.4	1.4	1.4
Temperature (K)			623.15	623.15
Is Main	yes	no	no	no
output type	**CH4**	**H2**	**axial air**	**axial air.1**
density	3.259283	0.409558	2.832036	2.832036
q	0.000254	0.000253	0.010657	0.010657
choke area	5.19E-07	1.83E-07	6.47E-06	6.47E-06
pressure pa	1083163	1056531	3098532	3098532
pressure psia	157.0995	153.2367	449.4039	449.4039
pressure psig	142.3995	138.5367	434.7039	434.7039
choked?	YES	YES	YES	YES

- Working under Professor Kareem Ahmed.
- Working on a script geared towards calculating emissions with respect to residence time using Cantera library.
- Studying the flashback phenomena of hydrogen fuel.
- Developed a Python condition calculator that reads json inputs and outputs standardized facility run parameters to reach choked-flow conditions and meet mission requirements.
- Assisted in the design and build of axial stage engine.

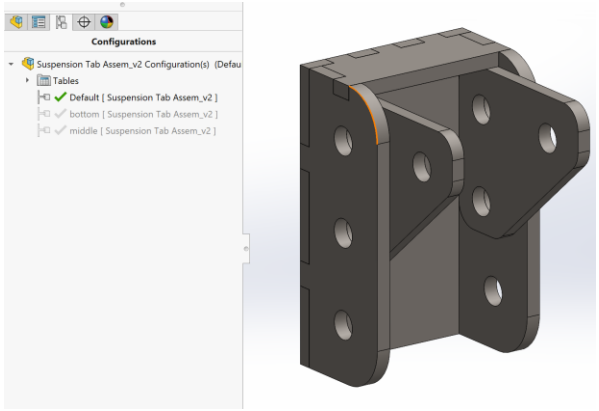
Experience | Knight Experimental Rocketry

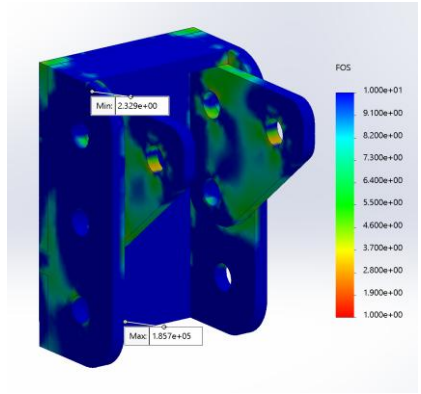
- Propulsion Lead: Applied basic rocketry principles (heat transfer, bolt stress, axial stress, thrust, and impulse) and implemented an effective layout of the system.
  - Delegated tasks to design and edit a preliminary design of the propulsion system.
  - Managed the master assembly for the propulsion system.
  - Built and tested propulsion chamber through static fires.
- Payload Team: Created a preliminary design for the payload frame, utilizing resources such as McMaster and NASA's 3u CubeSat to develop my ideas.
- Working towards obtaining L1 certification.

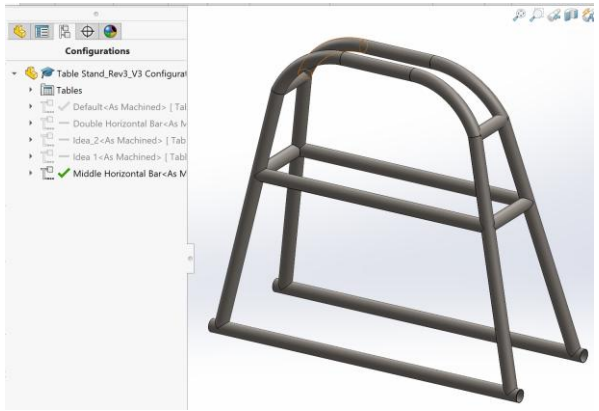


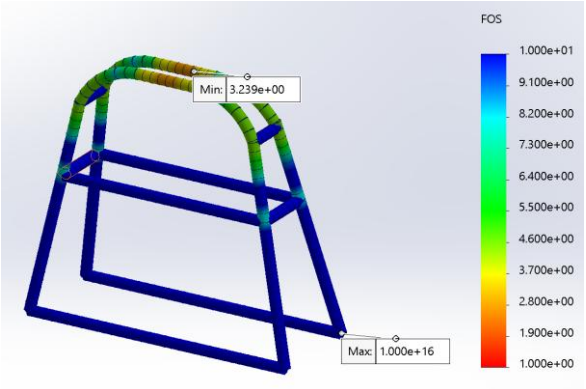
Experience | Knights Racing Baja SAE

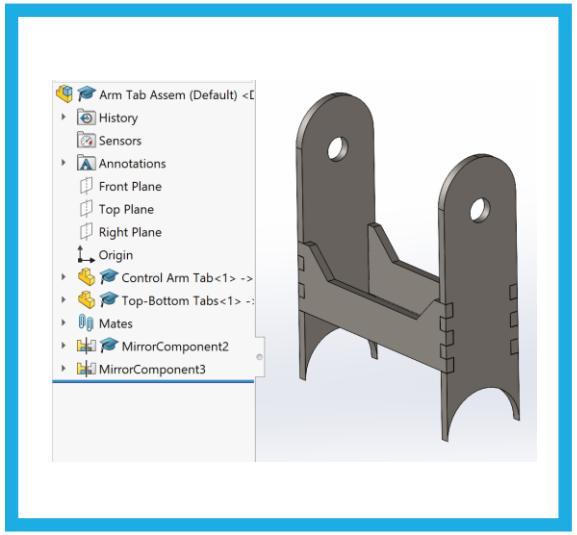
- Through small member projects, contributed to the overall design of the Baja SAE car for the 24’-25’ academic year.
  - Spacer for suspension
  - Suspension tabs for suspension
  - Stand to hold car up for maintenance
  - Arm Tabs
- Conducted FEA analysis on parts to determine if they were able to withstand forces given a certain material.

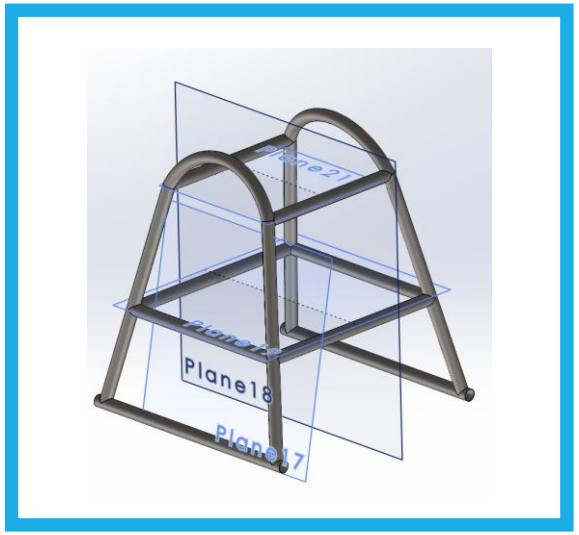


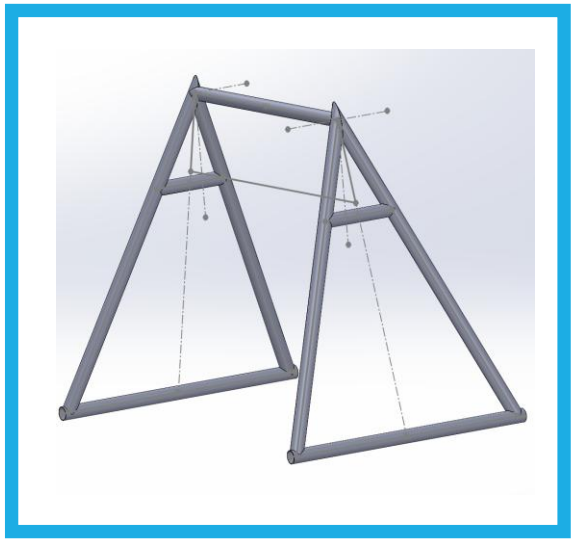


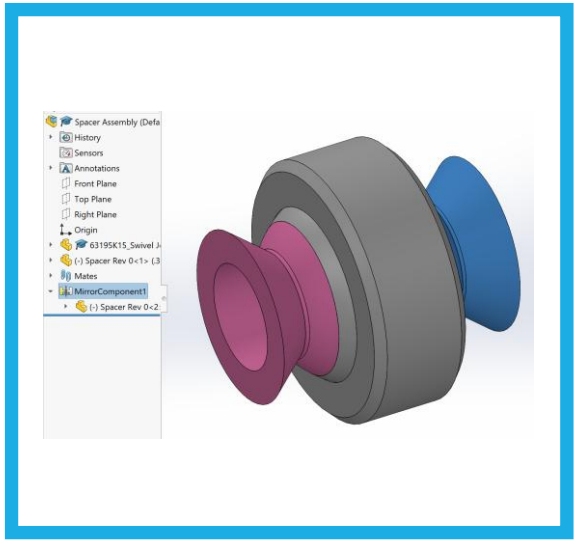








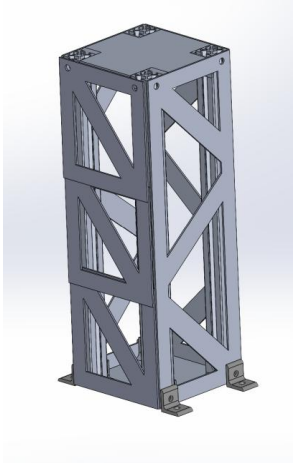




Experience | Knights Racing Baja SAE

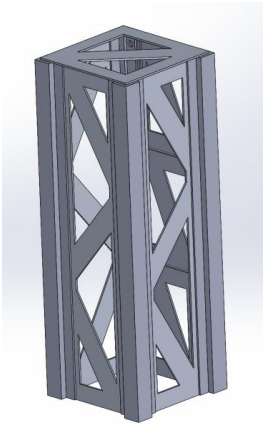
Experience | Society of Hispanic Professional Engineers

- Projects Committee – Payloads Team/Member
  - SHPE x KXR Collaboration: Designed a preliminary payload frame for the IREC 24’-25’ team.
  - Payload frame optimized storage/allowed for easy accessibility for internal components/provided stability.
  - Learned basic Arduino code and helped others understand basic concepts in SolidWorks.
- Professional Development – ResearchSHPE Co-Director
  - Coordinated events with my team to help students learn the importance of research and other opportunities.
- Outreach Committee – Volunteer Director
  - Reached out to organizations for volunteering opportunities.
  - Held meetings to discuss possible collaboration events
  - Participated in various outreach events.

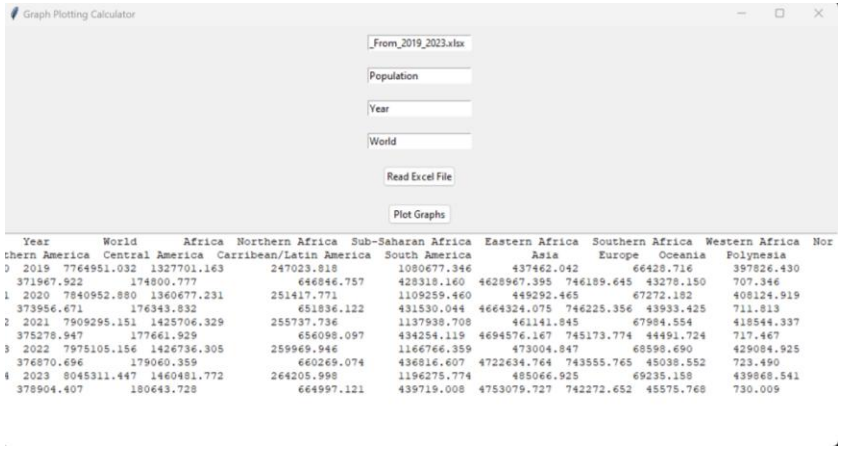


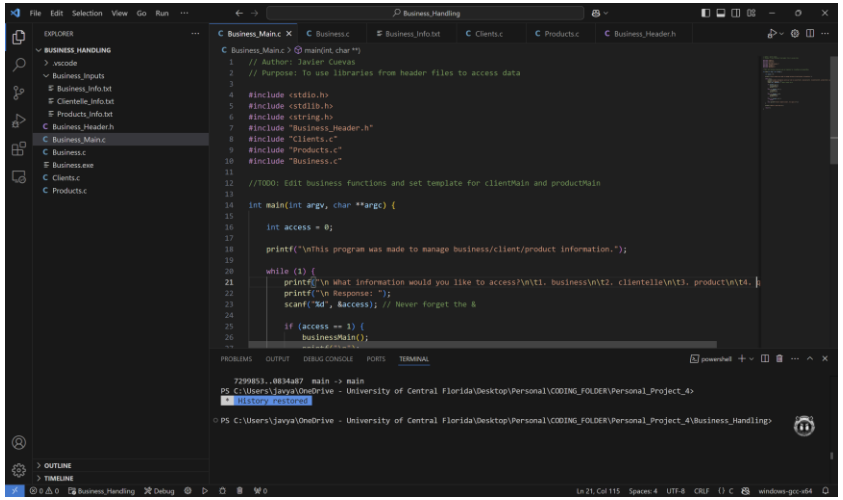






Experience | Personal Projects





- Made Python executable that would generate scatter plots based on Excel data.
- Developed C programs to strengthen foundational programming and logic skills, including one that calculates x-intercepts and y-values for quadratic functions and another that simulates a simplified dominoes game. These projects demonstrate problem-solving, algorithm design, and practical application of core C programming concepts.

Experience | Personal Projects

axis	y
y_i (m)	500
v_i (m/s)	100
g (m/s^2)(constant)	-9.8

displacement	
time (s)	4
p (x)	180
p (y)	321.6
p (z)	0
p (f)	368.5466

This will give you the distance from where you started to where you are currently. Otherwise known as displacement.

NOTE: In general physics, you probably don't have to worry about motion on the z axis, so keep v (z)/p (z) as 0 for now.

axis	x
x_i (m)	180
v_i (m/s)	45
a (m/s^2)(constant)	0

resulting velocity	
time (s)	8
v (x)	21.6
v (y)	45
v (z)	0
v (f)	49.915529

This will give you the resulting velocity based off of the velocity in the x, y, and z axis. Direction will be determined by you.

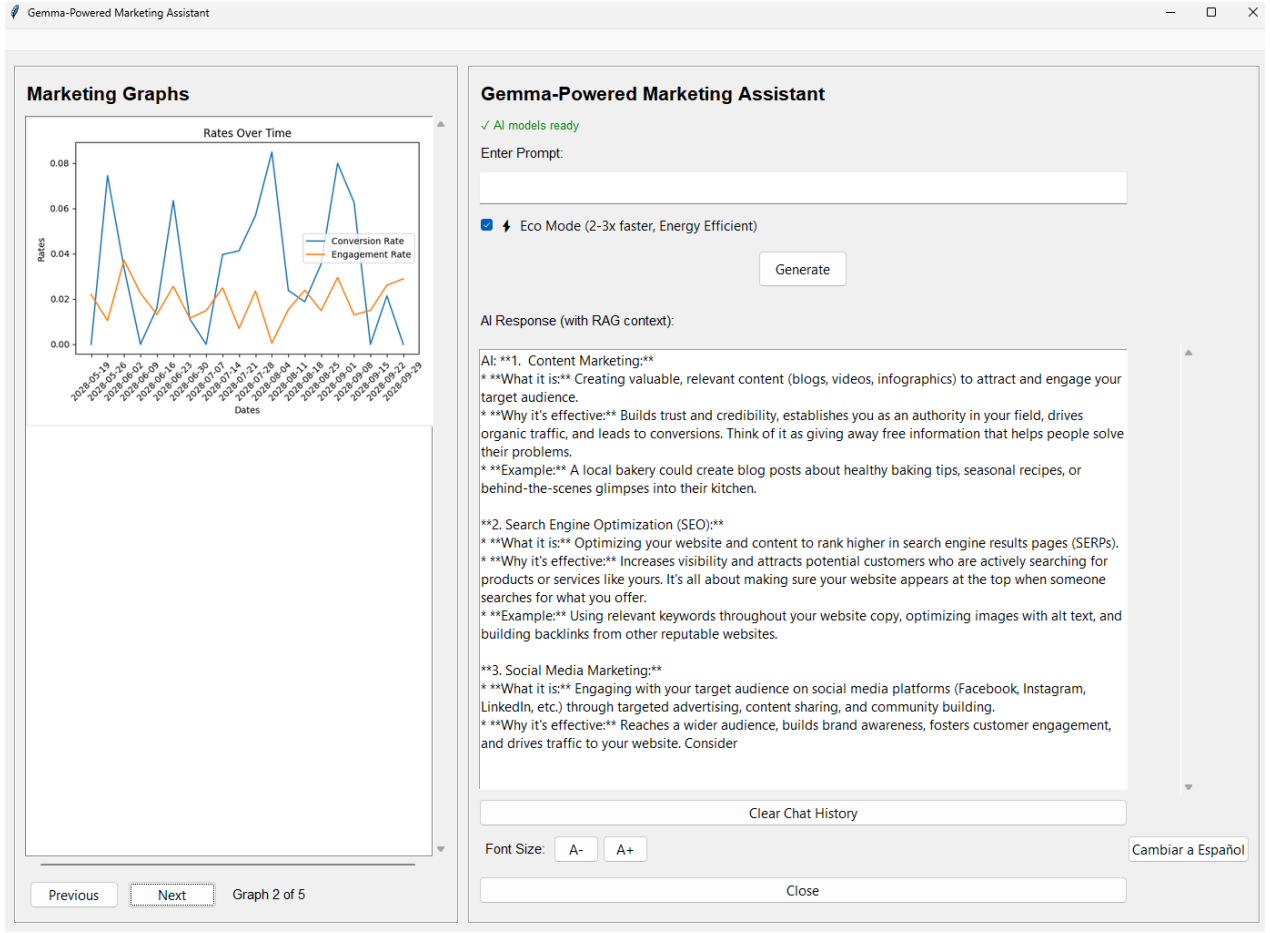
NOTE: When making these calculations, make sure that the units are correct and the direction is correct.

t (x)	position (x)	velocity (x)	acceleration (x)
0	180.000	45.000	0.000
2	270.000	45.000	0.000
4	360.000	45.000	0.000
6	450.000	45.000	0.000
8	540.000	45.000	0.000
10	630.000	45.000	0.000
12	720.000	45.000	0.000
14	810.000	45.000	0.000
16	900.000	45.000	0.000
18	990.000	45.000	0.000
20	1080.000	45.000	0.000
22	1170.000	45.000	0.000
24	1260.000	45.000	0.000
26	1350.000	45.000	0.000
28	1440.000	45.000	0.000

t (y)	position (y)	velocity (y)	acceleration (y)
0	500.000	100.000	-9.800
2	680.400	80.400	-9.800
4	821.600	60.800	-9.800
6	923.600	41.200	-9.800
8	986.400	21.600	-9.800
10	1010.000	2.000	-9.800
12	994.400	-17.600	-9.800
14	939.600	-37.200	-9.800
16	845.600	-56.800	-9.800
18	712.400	-76.400	-9.800
20	540.000	-96.000	-9.800
22	328.400	-115.600	-9.800
24	77.600	-135.200	-9.800
26	-212.400	-154.800	-9.800
28	-541.600	-174.400	-9.800

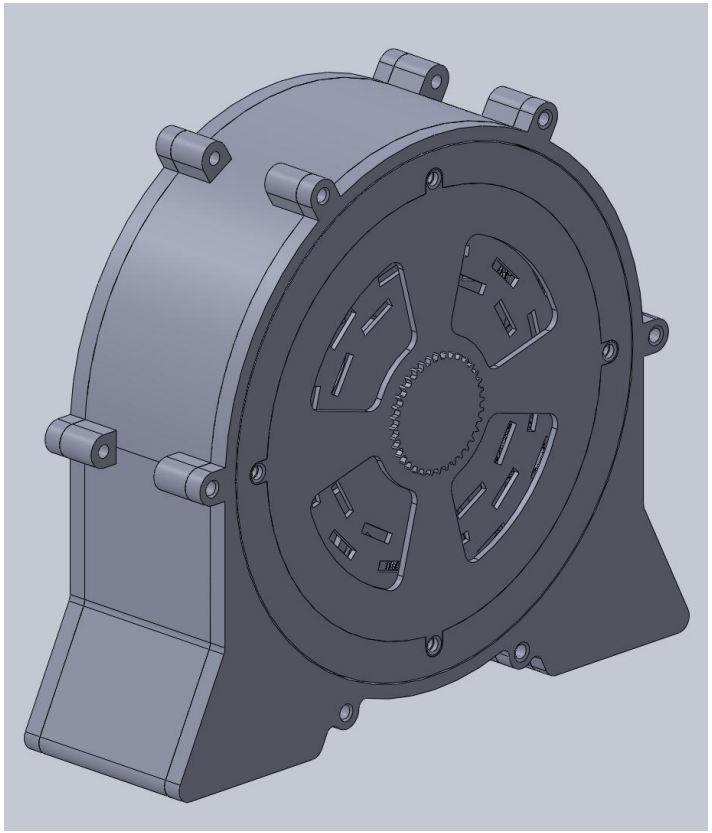
- Made Physics calculator for several different physics applications.
- Demonstrates basic physics concepts and integral/derivative processes.
- Utilized to help classmates understand Physics and Calculus concepts.

Experience | Personal Projects



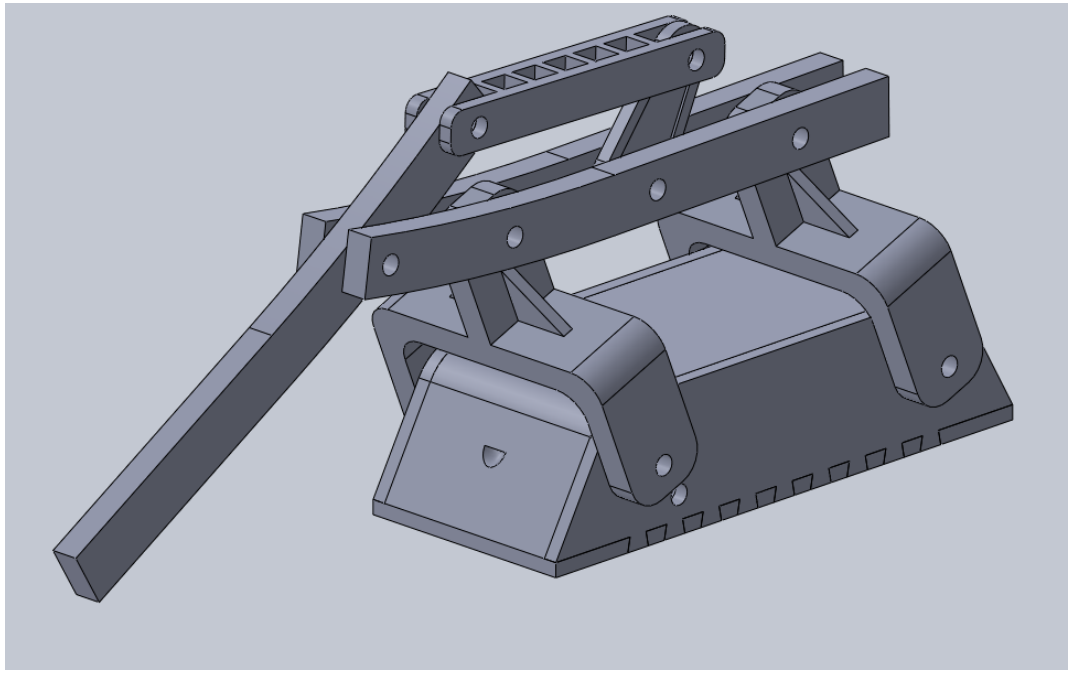
- Developed digital advertising AI software that theoretically could interpret market trends and ad performance, providing suggestions that would positively impact a businesses digital influence.
- Utilized Python Tkinter library with Google's Gemma AI model.
- Small Language Model designed to promote sustainability in a world dominated by resource-heavy Large Language Models.

Experience | Personal Projects



- Building an electric generator using concepts of electromagnetic induction.
- 3-D printed parts and components, used thin copper wire and small magnets to induce electricity from the wires into a battery.
- Design concept is to be implemented onto a mountain bike that will be turned into an electric bicycle.

Experience | Personal Projects



- Designing a robotic arm that can be remote controlled.
- Going through several iterations of possible robotic arm designs for maximum motor efficiency and control.
- Developed a housing to enclose the bottom motors and the Arduino uno R3 microcontroller.



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February 10, 2023
Date

Andrew Anagnost
President, Chief Executive Officer

anFn-s27N
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Certifications | Autodesk AutoCAD



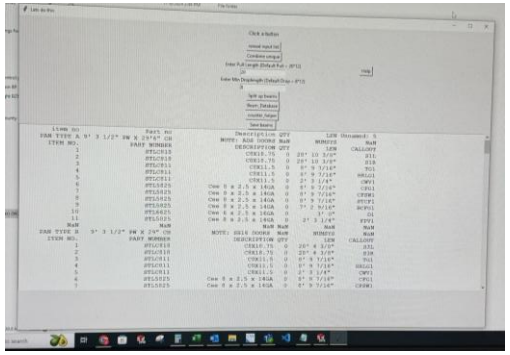
Certifications | CSWA

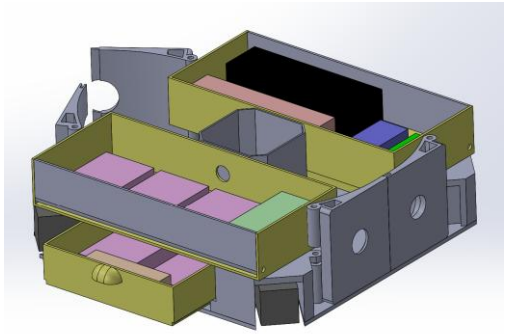


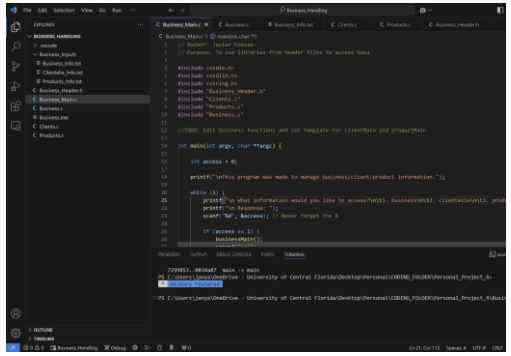
Certifications | CSWP

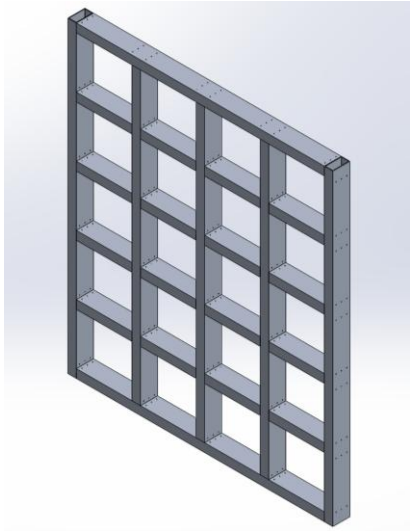
Certifications | Interim Security Clearance

Skills | Computer – Programming/CAD

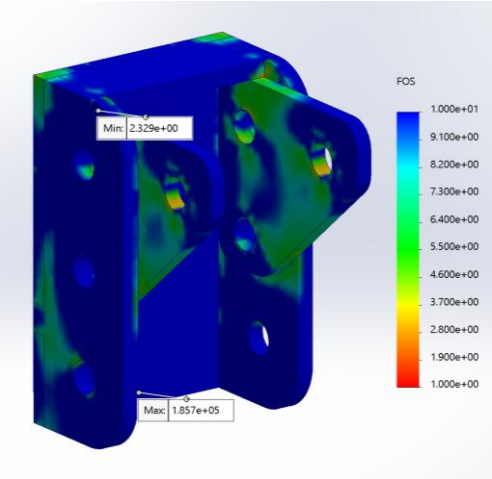


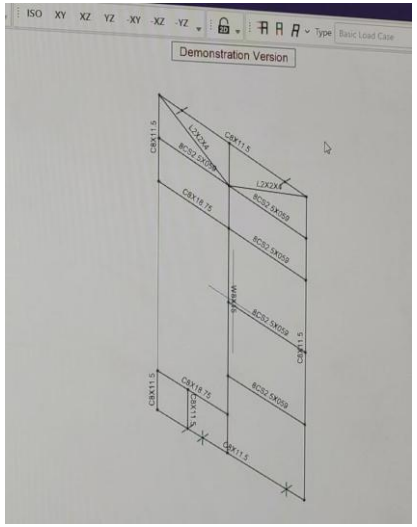


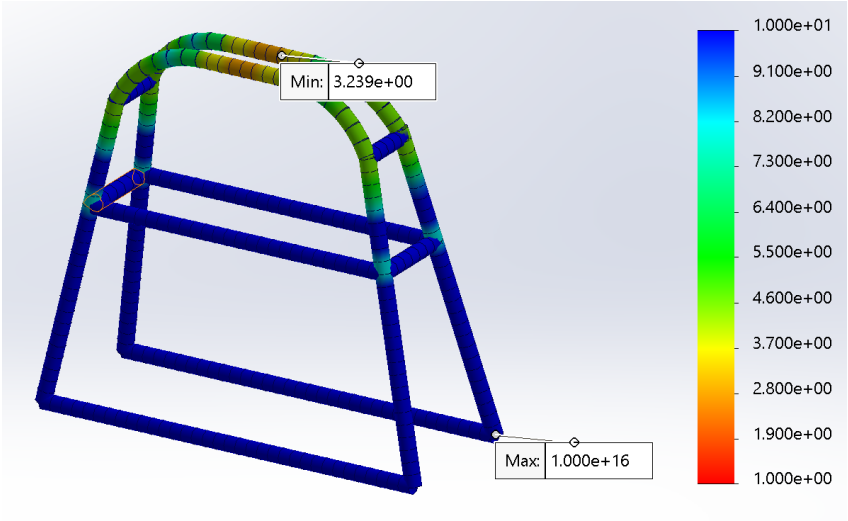




- SolidWorks: Expert, obtained CSWA/CSWP
  - Well Bilt Industries, KXR, SHPE, Baja, personal projects
- Fusion 360: Beginner
- Autodesk: Proficient, obtained AutoCAD certification
  - West Port High
- Programming: Proficient in Python, C, MATLAB, Java, HTML.
  - Leveraged Python to create an executable file that would organize beam cut lists into separate beams for shipping purposes.
  - Using Python for Engineering calculations. Using Tkinter library to design applications for versatile use in Aerospace field.
  - Utilized C to create executable programs that would help users manage business/client/product information. Created several other programs for faster calculations.





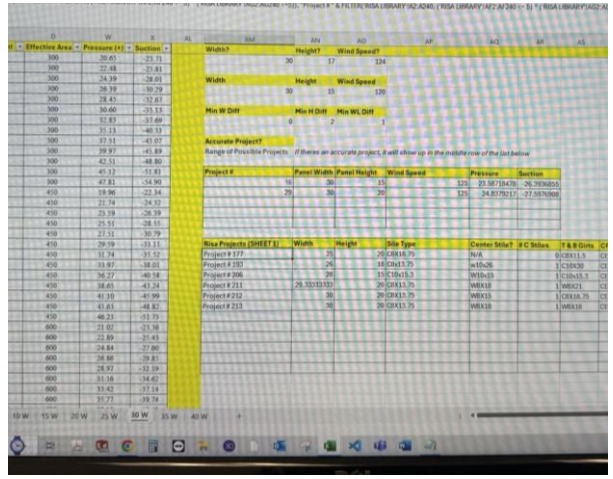


Skills | Computer – FEA Analysis

- Ansys: Learning
- SolidWorks Studies: Proficient
  - Utilized to make calculations on suspension tabs, stands, spacers.
  - Used to determine the best geometry for robotic arm.
- RISA – Structural Analysis: Proficient
  - Utilized in Well Bilt Industries to calculate wind loads on door panels.
  - Determined what beam types/configurations/profiles (I-beams, hot-rolled stiles and cold-formed girts, tubes for hydraulics, etc.) are most optimal given certain wind speeds

Skills | Computer - Microsoft

- Microsoft Excel: Expert, utilized in previous company and school settings
  - Well Bilt Industries, KXR, SHPE, PREM Scholars, personal calculations/schedules
  - Utilized to demonstrate basic physics concepts
  - Able to translate from Excel to Google Sheets
- Microsoft Word: Proficient
  - West Port High, College of Central Florida, KXR
- Microsoft PowerPoint: Proficient, utilized to demonstrate preliminary designs
  - West Port High, College of Central Florida, KXR
- Microsoft Teams: Proficient, good form of communication and utilized it's services to conduct meetings.
  - West Port High, College of Central Florida, KXR, SHPE, PREM Scholars





Skills | Manufacturing

- Computer:
  - Beamline: Proficient
    - Utilized to manufacture beams and ship them out
  - Messer Nesting: Proficient
    - Utilized in Well Bilt to cut out plates of certain thickness
    - Utilized SolidWorks to get the 2-D configuration of these plates
- Soldering: Beginner

Skills | Languages

- Spanish: Proficient, learned Spanish since birth
  - Utilized in Well Bilt industries to communicate ideas between clients and coworkers
- English: Proficient, second language and is always practiced
  - Common language utilized to communicate with others
- French: Beginner, learning slowly and will soon use it to learn other languages.

Links

Handshake | Javier Cuevas Chabrier

- https://app.joinhandshake.com/profiles/56146496?ref=students

GitHub | Javy0119

- https://github.com/Javy0119

LinkedIn | Javier Cuevas

- www.linkedin.com/in/javier-cuevas-JACC84

Portfolio

- https://javy-scratchspace.github.io/javy0119_portfolio.io/